Chapter 1

Snow basics

Snow

1.1 Basic properties of snow

For the atmosphere to release its water content, whether in liquid or solid form, the air must first reach a state of supersaturation with respect to water or ice. This condition is typically achieved through upward motion of air masses which causes adiabatic cooling of air parcels. This temperature drop strongly reduces air's capacity to hold water vapor, driving the parcel toward, and eventually beyond, saturation (Wallace and Hobbs 2006).

Snowfall can only be produced by clouds that extend above the 0 °C level. Although temperatures within these cold clouds can drop well below freezing point, supercooled water droplets can still coexist with ice particles. The freezing process, known as ice nucleation, occurs through two different mechanisms:

- *homogeneous nucleation*, which occurs in pure water droplets without foreign particles. The first step in the freezing process is the formation of an ice embryo, which must exceed a critical size to produce a decrease in the free energy of the system. Under these conditions, the freezing process continues until the entire water droplet is turned into ice. The average size of the ice embryo increases as the temperature decreases, so that below a certain threshold temperature, which is around −40 °C in this case, freezing becomes almost certain. If homogeneous nucleation were the only mechanism responsible for ice particle formation, snowfall would be extremely rare at mid-latitudes. Since this is not the case, an additional mechanism is required to increase the threshold temperature.

- *heterogeneous nucleation*, which relies on the presence of a foreign particle.

Cloud droplets typically contain such impurities acting as freezing nuclei. Their presence promotes the formation of an ice-like structure, thereby enabling freezing at temperatures considerably higher than those required for homogeneous nucleation. It is also possible for foreign particles not to be part of the water droplet itself, but instead to trigger freezing through contact or deposition. Regardless of the specific mechanism involved, the most effective ice nuclei are insoluble in water and have a crystallographic structure similar to ice.

Once nucleated, the ice particle grows through further microphysical processes, whose relative dominance determines its final shape and size. The three main crystal growth mechanisms are:

- *vapor deposition*, which occurs whenever the air is supersaturated with respect to ice. For example, in mixed-phase clouds dominated by supercooled droplets, vapor molecules are deposited onto the ice crystal, causing it to grow significantly faster than the surrounding water droplets. The growth rate reaches a maximum around $-14\,°C$, since at that temperature the difference between the saturated vapor pressure over water and over ice is largest, thus driving vapor molecules towards ice crystals. Studies on the growth of ice crystals by deposition have shown a strong relationship between crystal shape and the temperature at which the growth process takes place. This strong temperature dependence produces an alternating pattern between plate-like and column-like ice particles, as shown in Figure 1.1.

- *riming*, which occurs exclusively in mixed-phase clouds. Whenever an ice crystal collides with a supercooled droplet, the latter may freeze onto the former, resulting in an increase in ice mass. The collision frequency depends on cloud density and composition, but typically numerous riming events take place, sometimes producing a graupel, an ice particle in which the original ice crystal can no longer be identified. Riming can also lead to the formation of hailstones, which typically range from a few millimeters to several centimetres in diameter.

- *aggregation*, caused by collisions between ice particles. The probability of adhesion is strongly linked to crystal shapes and increases with increasing temperature, reaching a maximum at around $-5\,°C$, where ice surfaces become particularly "sticky".

Once an ice particle grows large enough to overcome updraft forces and turbulent motion within the cloud, it begins to fall toward the ground. The temperature

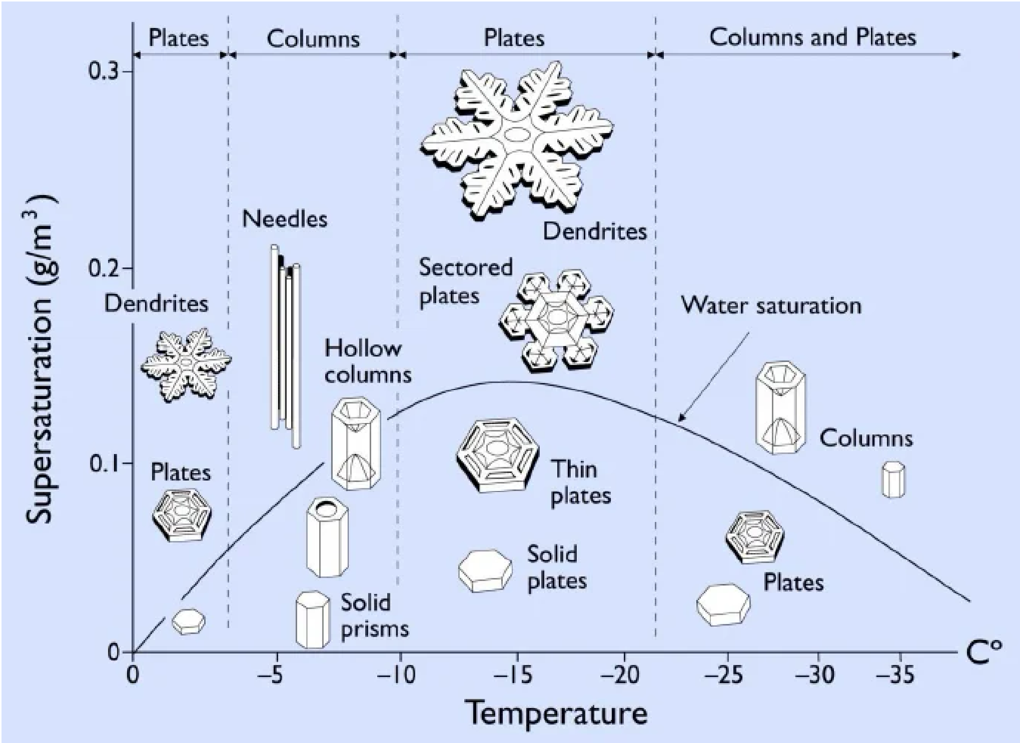

Figure 1.1: First try

profile encountered by the descending particle ultimately determines the type of precipitation that reaches the surface. For snowfall to occur, the snowflake, composed of up to hundreds of ice crystals, must remain predominantly below freezing and avoid melting completely before reaching the ground. Typically, snowflakes have fall rates of approximately $1.5\,\mathrm{m\,s^{-1}}$. Snowfall amounts are measured either by depth or by snow water equivalent (SWE), defined as the height of the water column resulting from a complete melting of the snow. Snow depth typically increases of approximately $1\,\mathrm{cm\,h^{-1}}$ during snowfall events. Newly fallen snow density ranges from $20\,\mathrm{kg\,m^{-3}}$ to $300\,\mathrm{kg\,m^{-3}}$ depending on wind patterns and weather conditions, with higher temperatures generally leading to higher density values (Armstrong and Brun 2008).

1.1.1 Snow metamorphism and compaction

Accumulation of snowflakes on bare ground or atop older snow results in the formation of a new snow layer. Snow crystals form a solid matrix delimiting pores filled with humid air or liquid water or both, depending on layer temperature. This structure can be clearly appreciated in Figure 1.2, where the coexistence of the solid, liquid and gaseous phases of water is visible. Snow is thus a porous medium, whose properties are closely linked to its texture. The arrangement of snow crystals changes over time

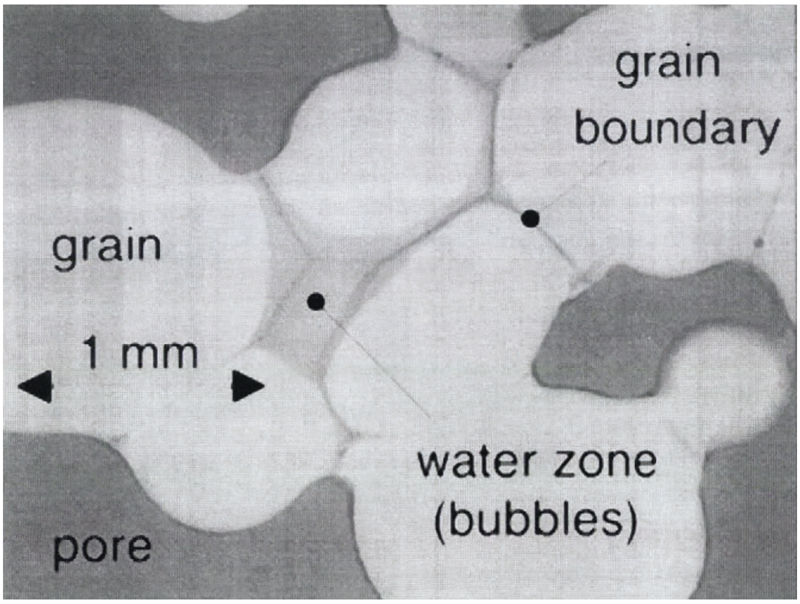

Figure 1.2: Image from Brzoska et al. 1998

primarily as a result of the thermodynamic relationships among the constituents of the snow layer. This process, known as *snow metamorphism*, depends strongly on thermal and meteorological conditions. A key factor governing ice particles evolution is the presence or absence of liquid water, which motivates the distinction between dry and wet snow metamorphism. The former occurs when pores are filled only by air, typically saturated with water vapor. In this case, the main driving factor are the variations in crystal surface curvature and temperature, which induces local differences in equilibrium vapor pressure. The main mechanisms underlying dry snow metamorphism are:

- *equilibrium growth*, driven by crystal shape. Local variations in surface curvature generate corresponding variations in equilibrium vapor pressure, inducing local pressure gradients that drive vapor from convex toward concave (or less convex) regions. Over time this transfer, sustained by sublimation on convex areas and deposition on concave ones, rounds ice grains, driving snow crystals toward a spherical shape.

- *kinetic growth*, related to the presence of a temperature gradient. Because saturation vapor pressure depends strongly on temperature, such a gradient causes vapor to diffuse from the warmest crystals toward the coldest ones. Here too, sublimation on warm crystal and deposition on cold ones sustain the process, which, when the temperature gradient is strong enough, leads to the formation on facets.

In practice, these two mechanisms are rarely mutually exclusive: they act simultaneously within the snowpack, with their relative strengths determined mainly by the magnitude of the local temperature gradient compared to the grain's curvature effect. In particular, the kinetic growth of snow particles is a matter of considerable concern in avalanche forecasting, since faceted crystals, if the temperature gradient persists over time, can evolve in depth hoar. This type of layer is notoriously weak and prone to failure, often resulting in large-scale avalanches. When liquid water is present within the snowpack, the dynamics change substantially: mass exchanges between the three water phases become essential to understand wet snow metamorphism. The main driver of ice particles evolution is, once again, crystal surface curvature, with convex areas associated with lower melting point temperatures. This topological property leads to the melting of the most convex surfaces and to refreezing on the more concave ones, driving ice particles towards the equilibrium spherical shape. The efficiency of this metamorphism channel is linked to the freedom water has to move within the snowpack: at very low water content, capillary effects are negligible, and mass exchanges between the solid and gaseous phases may still dominate.

Regardless of the specific metamorphic process involved, the evolution of ice crystals ultimately results in the compaction of the snow layer. Wind also plays a crucial role in increasing snow density, primarily through the fragmentation of snowflakes upon collision, which allows for a tighter packing of ice particles. Finally, gravity contributes to snow compaction as well, since buried snow layers must withstand an increasing load exerted by the weight of the overlying layers, resulting in a higher deformation layer. The combined effects of metamorphism and deformation strain are typically described through a Newtonian viscosity approach. Snow viscosity depends on several parameters, including temperature, liquid water content and porosity and its values are generally retrieved by means of empirical relations. Overall, the density of older snow can easily exceed values of $400\,\mathrm{kg\,m^{-3}}$, sometimes even reaching $500\,\mathrm{kg\,m^{-3}}$ in specific locations, depending on water content and time of the year (Jonas et al. 2009).

1.1.2 Snowpack energy balance

1.2 Snow measurement methods

Knowledge of snow cover properties is essential for assessing solid water storage in the snowpack, a critical factor influencing agriculture and hydroelectric power generation, as well as for avalanche risk evaluation.

Historically, snow metrics were recorded manually, requiring an operator to conduct manual measurements on a daily basis whenever feasible. To ensure continuous data collection even in remote areas, automated weather stations (AWS) have been introduced, and are steadily replacing manual ones as well. An automated monitoring network provides a comprehensive overview for hazard assessment and for studying snow dynamics over decades. However, despite the significant number of AWS currently in operation, achieving high-resolution coverage remains challenging due to high costs and maintenance issues. Consequently, certain data can only be acquired via remote sensing, by means of Earth-observing satellites.

While remote sensing provides spatially distributed data across entire regions, manual and automated measures are taken at specific locations, which are chosen according to strict criteria. To ensure the reliability of these in situ observations, the monitoring site must match the features of the surrounding terrain. In alpine regions, monitoring sites must be clear of large boulders and protected from excessive sun exposure, which can cause premature snowmelt and yield unrepresentative data. It is also important to avoid locations prone to preferential snow accumulation driven by local wind patterns.

These guidelines represent only the baseline requirements for reliable snow measures. A more comprehensive analysis of the operational details is provided in the following sections.

1.2.1 Manual measures

Manual snow depth (HS) records are an effective tool for assessing long-term trends, as they can span over several decades. Their longevity is due to a straightforward measuring procedure, requiring only a stake or an extendible graduated rod.

The snow stake should be vertical, painted white to minimize melt and fixed to the ground before accumulation begins. It is required to make snow depth observations from a sufficient distance to avoid any deformation of the snowpack. Because the stake acts as an obstacle to wind and snow redistribution, localized mounds can form around it. Under such circumstances, operator must visually estimate the undisturbed snow surface to recover the actual HS value.

For a non-fixed observation, the snow depth value should be evaluated at the same location each time. The observer needs to ensure that the device, manually inserted in the snowpack, reaches the base surface without penetrating any soil thus overestimating the actual HS. The rod has to be vertical, and an estimate of the slope angle should be paired to every measure.

Closely linked to snow depth is snowfall depth (HN), which can only be obtained manually. To avoid wind-induced bias this measurement should be carried out in a sheltered location, so that snow accumulates undisturbed on an artificial surface for the prescribed interval (usually 24 hours). Just before clearing the surface, thereby starting the next observation period, a ruler is inserted vertically into the freshly formed layer to obtain a depth measurement.

Although HS and HN define the vertical extension of the snowpack, hydrological applications rely primarily on the snow water equivalent (SWE). As shown in (cite equation), SWE is directly linked to snow depth through the bulk density of the snowpack. Although various manual measurement techniques exist, this section focuses on the two primary methods, snow pits and snow tubes.

A snow pit consists of a manually-dug hole up to a reference surface, usually the ground. Initially, the observer performs a stratigraphic analysis to deepen his knowledge of the internal structure of the snowpack. The actual SWE measurement is conducted by inserting a sampler of known volume perpendicularly into the exposed snow face; the sample is then weighted to assess snow density. This procedure is usually repeated at multiple depths, either for each identified layer or at fixed intervals, such as every 20 centimetres. This method enables the observer to reconstruct the snow density profile which, when combined with depth information, yields both the layer-by-layer and total snow water equivalent at that specific location. Clearly, the snow pit technique is disruptive and needs to be performed at a different site each time.

If the measurement campaign is focused on a spatial survey rather than a detailed, locally defined inspection, the logical choice is utilizing a snow tube, because it allows for rapid sampling of the entire snow column to directly retrieve the total SWE, sacrificing layer-specific information. The snow tube is inserted vertically in the snowpack until it reaches the ground. An estimate of the extracted core volume is made, and the average density is then computed by weighting the snow column. The conversion to SWE is then trivial. This device should be used where the snow water equivalent is less spatially homogeneous, such as locations with heterogeneous exposure to both wind patterns and solar radiation.

1.2.2 Automatic measures

Automatic weather station (AWS) networks do not require manual intervention, except for routine maintenance and instrument calibration. An AWS provides continuous observations even in remote locations where human operation would be unfeasible. Typically, an alpine automatic weather station is equipped with a variety of sensors to provide a highly comprehensive overview of environmental conditions.

Automated HS measurements can be conducted using instruments based on either sonic or optical (laser) technology. Sonic sensors transmit an ultrasonic pulse towards the snow surface and detect the returning echo. Since the speed of sound is temperature-dependent, a temperature reading is also required to ensure accuracy. On the other hand, laser instruments determine the target surface distance by analyzing signal interference. These measurements are actually much more precise, but they come at a higher energy cost. In both scenarios, the sensor measures the distance to the upper layer of the snowpack, from which the snow depth is then retrieved

$$\text{HS} = \text{Zero depth distance} - \text{Distance to target} \tag{1.1}$$

The zero depth distance must be determined before snow accumulation begins, and is a key part of instrument calibration. To obtain reliable measurements, the HS sensor should be installed at a sufficient height above the maximum anticipated snow depth to prevent burial during winter, which would result in data loss. In addition, minimizing interference with the support structure is essential, as the mast acts as a wind obstacle and can induce measurement artifacts. Consequently, HS sensors are typically installed on a horizontal arm that stretches outward, ensuring that the target area remains at a sufficient distance from the mounting pole.

SWE monitoring can also be automated; to this end, several techniques have been developed, the most common of which relies on snow pillows. This instrument consists of a rubber bladder filled with an antifreeze liquid mixture and coupled with a pressure transducer, which provides the SWE estimate. The snow pillow is installed flush with the ground surface, thereby supporting the weight of the entire snowpack. As snow accumulates on top of the pillow, the hydrostatic pressure increases and this variation is transduced into snow water equivalent values. Snow pillows were first introduced in the late 1960s, and have proved to be a useful tool for SWE measurements (Beaumont 1965). However, they are currently being replaced by snow scales due to the toxicity of the antifreeze fluid, which poses an environmental hazard to the installation site.

1.2.3 Remote sensing

Satellite remote sensing (SRS) serves as a useful tool to fill critical gaps in current knowledge regarding climatic variables, enabling researchers to monitor, usually on a daily basis, even remote locations which could not be accessible by a human operator. Furthermore, SRS data substantially improve meteorological reanalysis products, which constitute a cornerstone of contemporary climate change research. Earth observing satellites can be classified into two distinct categories based on their orbital characteristics:

- Polar orbit satellites, which travel roughly from north to south over the poles, targeting a specific longitudinal strip per pass. The entire globe is thus mapped over consecutive orbits. Usually, these satellites operate at an altitude of approximately 700 km, completing a full revolution in under two hours.

- Geostationary satellites, which occupy a geosynchronous orbit, thereby allowing for real-time meteorological monitoring on a specific target area. Their altitude of approximately 36000 km above Earth's equator results in a significantly lower spatial resolution compared to that of polar orbit satellites.

The choice of orbit is strictly driven by observational priorities. Consequently, satellites dedicated to cryospheric research favor polar orbit, as maximizing spatial resolution is essential to capture the highly variable distribution of snow coverage across complex terrains.

Satellite-based observations of Snow Cover Extent (SCE) began in 1967, through the efforts of National Oceanic and Atmospheric Administration (NOAA). This monitoring program focused on the Northern Hemisphere, where most of the world's terrestrial snow cover is found, with observations conducted on a weekly basis. Despite the preliminary nature of this work, it highlighted the strong latitudinal dependence of snow cover duration and paved the way for more comprehensive analyses. One of the main limitations of these early observations was their coarse spatial resolution of $\sim 200\,\mathrm{km}$, which was insufficient to adequately characterize snow cover in many mountainous regions (Bormann et al. 2018).

The next big leap in sensor technology came with the introduction of the Advanced Very High Resolution Radiometer (AVHRR), which measures Earth radiance in five spectral bands to obtain information on temperature, albedo and cloud coverage distribution. The first tests on a spaceborne sensor took place in 1978, but reliable snow cover data are available only from 1981 onward. This snow product has been used to estimate snow cover duration and trends since the middle eighties in different mountainous regions such as the Alps, where SRS data highlight significantly negative SC trends throughout the 1990s, followed by a partial recovery at the beginning of the new millennium (Hüsler et al. 2014). Although AVHRR offers a significant improvement in spatial resolution with its pixels of $\sim 1\,\mathrm{km}$, data reliability is limited by the presence of only one thermal channel for snow-cloud discrimination (Fugazza et al. 2021).

To overcome these limitations, the Moderate Resolution Imaging Spectroradiometer (MODIS) was launched in 1999 aboard the Terra satellite, followed by a second unit on Aqua in 2002. Both operate in polar orbit, jointly covering the whole globe every one to two days. MODIS is a 36-band spectral sensor spanning from the visible (0.41 µm) to the long-wave infrared (14.4 µm), with a spatial resolution of 500 m. To detect the presence of snow on the ground, the Normalized Difference Snow Index (NDSI) is used, defined as

$$\mathrm{NDSI} = \frac{\mathrm{G} - \mathrm{SWIR}}{\mathrm{G} + \mathrm{SWIR}}, \tag{1.2}$$

where G and SWIR stand for the green and shortwave infrared spectral bands. This normalized difference is close to one for a snow-covered pixel, since snow exhibits very high reflectance in the visible part of the electromagnetic spectrum, whereas it strongly absorbs radiation in the SWIR. An NDSI value greater than zero indicates the presence of snow in the pixel; in practice, a threshold of 0.4, corresponding to roughly 50 % snow coverage, is commonly adopted to classify a pixel as snow covered (Salomonson and Appel 2006).

Although MODIS has proven to be a robust and well-validated tool for snow cover characterization, the snowpack information it provides remains partial, as it is limited to detecting the presence or absence of snow. In order to assess how much water is stored in solid form on the ground, three main spaceborne techniques have been developed: passive microwave radiometry, gravimetric data and Synthetic Aperture Radar (SAR).

Passive microwave radiometry (PMW) is by far the most common method to remotely evaluate SWE. Given that the snowpack causes attenuation to the microwaves emitted by the Earth, comparing signal intensities between two frequencies ($\sim$ 18 GHz, and $\sim$ 37 GHz) makes it possible to estimate snow water equivalent. This is the only spaceborne technique that offers the possibility of long-term studies because global daily maps have been available since the launch, in 1978, of the Scanning Multichannel Microwave Radiometer (SMMR) by NASA (Schilling et al. 2024). Even though the temporal coverage of these series is significantly better than that of other available sources, several factors have been shown to influence data reliability, such as signal saturation caused by a deep snowpack.

As an alternative to radiation based measurements, gravimetric methods offer a different approach that avoids signal saturation by focusing on mass changes.

The accumulation of snow on the Earth's surface causes variations in the Earth's gravitational field, which can be converted into SWE estimates. This procedure is mainly employed by the Gravity Recovery and Climate Experiment (GRACE) mission. Due to its global coverage, this SWE dataset features a coarse spatial resolution of roughly 300 km, which remains useful for continental-scale hydrological studies (Schilling et al. 2024).

While passive microwaves (PMW) and gravimetric techniques are passive remote sensing methods, Synthetic Aperture Radar is an active remote sensing technology that transmits electromagnetic waves and records their backscattered echoes from the Earth's surface. Due to its active nature, SAR does not rely on solar illumination and can penetrate cloud cover, enabling data acquisition under every meteorological condition. Snowpack properties, such as SWE and snow grain size can be estimated remotely, as longer SAR wavelengths are even capable to penetrate into the snowpack. Furthermore, the phase information recorded by SAR can be exploited to monitor ground deformation and assess the stability of ground features. Although SAAR provides spatial resolution on the order of tens of metres, its temporal resolution is significantly lower than that of PMW and gravimetric methods, as the revisit time of a specific location is typically five days (Tsai et al. 2019).

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